

Sürekli-zamanlı Konvülyasyon (Continuous-Time Convolution) Hafta5

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Textbook: Linear Systems and Signals, B.P. Lathi

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Convolution Integral

- Commonly used in engineering, science, math

$$f_1(t) * f_2(t) \equiv \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

- Konvülasyon Özellikleri (Convolution properties)

- Yer Değiştirme (Commutative): $f_1(t) * f_2(t) = f_2(t) * f_1(t)$
- Dağılma (Distributive): $f_1(t) * [f_2(t) + f_3(t)] = f_1(t) * f_2(t) + f_1(t) * f_3(t)$
- Birleşme (Associative): $f_1(t) * [f_2(t) * f_3(t)] = [f_1(t) * f_2(t)] * f_3(t)$
- Kayma (Shift): If $f_1(t) * f_2(t) = c(t)$, then
 $f_1(t) * f_2(t - T) = f_1(t - T) * f_2(t) = c(t - T)$.
- Convolution with impulse, $f(t) * \delta(t) = f(t)$
- Convolution with shifted impulse, $f(t) * \delta(t - T) = f(t - T)$

important later in modulation

Graphical Convolution Methods

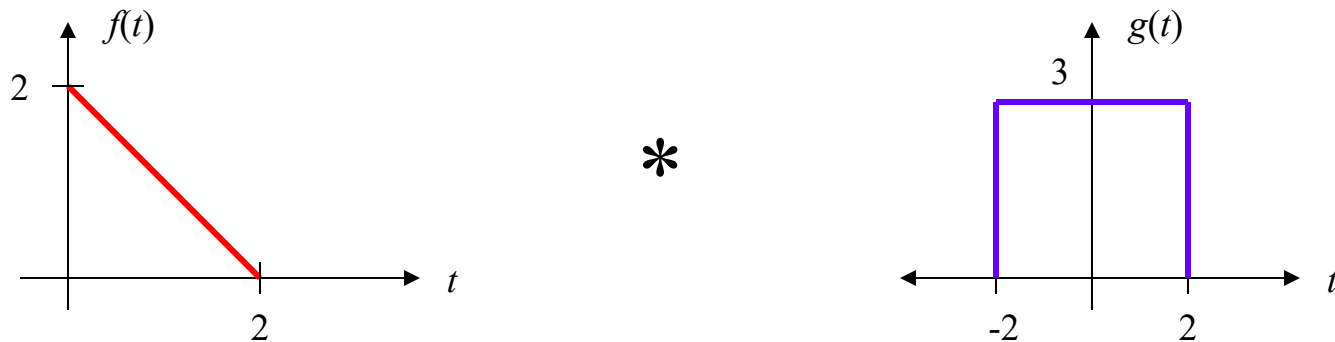
- From the convolution integral, convolution is equivalent to

$$f_1(t) * f_2(t) \equiv \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

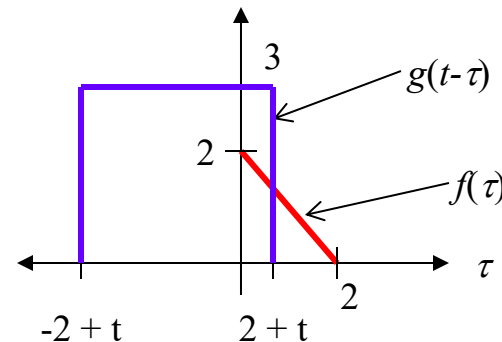
- Rotating one of the functions about the y axis
- Shifting it by t
- Multiplying this flipped, shifted function with the other function
- Calculating the area under this product
- Assigning this value to $f_1(t) * f_2(t)$ at t

Graphical Convolution Example

- Convolve the following two functions:



- Replace t with τ in $f(t)$ and $g(t)$
- Choose to flip and slide $g(\tau)$ since it is simpler and symmetric
- Functions overlap like this:

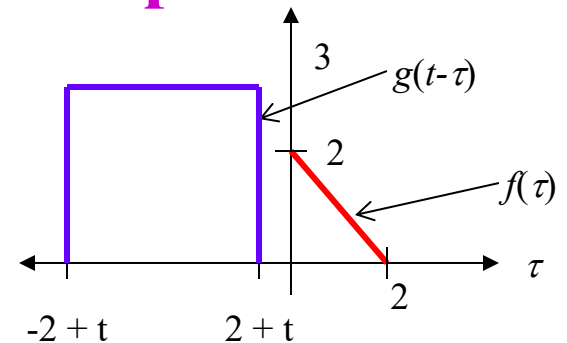


Graphical Convolution Example

- Convolution can be divided into 5 parts**

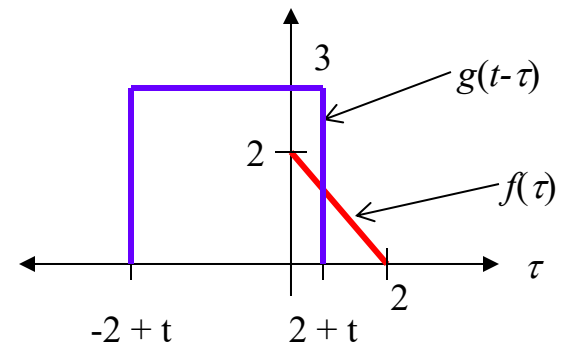
I. $t < -2$

- Two functions do not overlap
- Area under the product of the functions is zero



II. $-2 \leq t < 0$

- Part of $g(t)$ overlaps part of $f(t)$
- Area under the product of the functions is



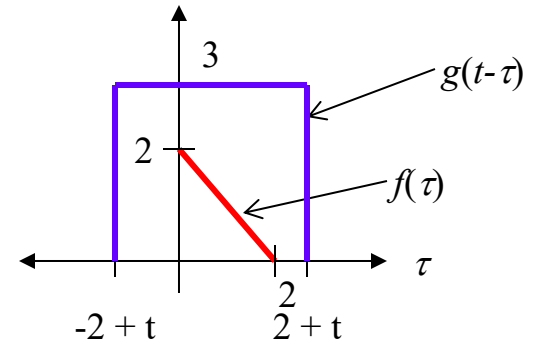
$$\int_0^{2+t} 3(-\tau + 2) d\tau = 3 \left(-\frac{\tau^2}{2} + 2\tau \right) \Big|_0^{2+t} = -\frac{3(2+t)^2}{2} + 6(2+t) = -\frac{3t^2}{2} + 6$$

Graphical Convolution Example

III. $0 \leq t < 2$

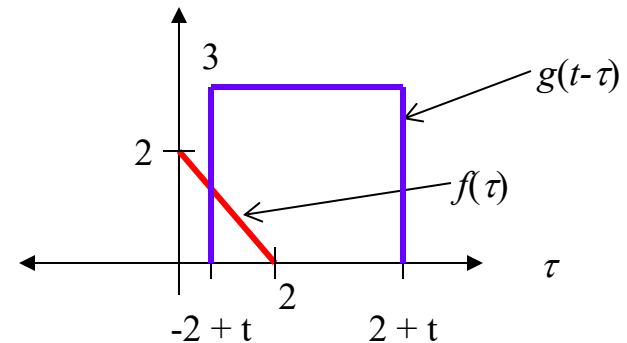
- Here, $g(t)$ completely overlaps $f(t)$
- Area under the product is just

$$\int_0^2 3(-\tau + 2) d\tau = 3 \left(-\frac{\tau^2}{2} + 2\tau \right) \bigg|_0^2 = 6$$



IV. $2 \leq t < 4$

- Part of $g(t)$ and $f(t)$ overlap
- Calculated similarly to $-2 \leq t < 0$



V. $t \geq 4$

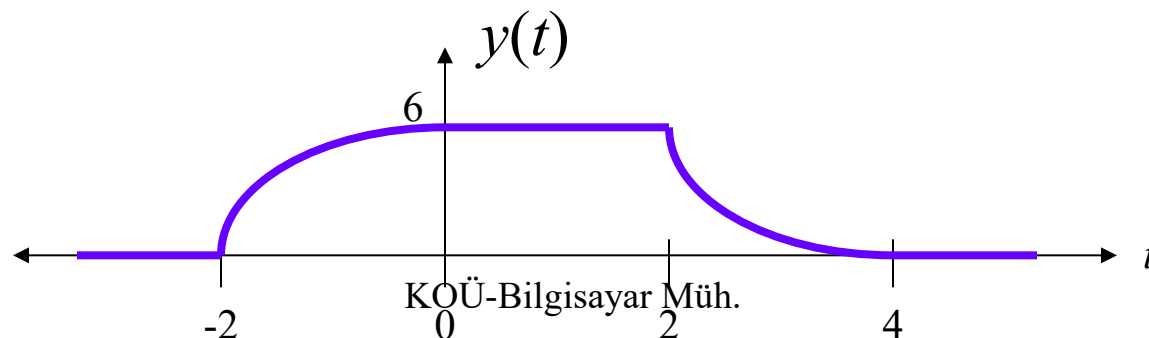
- $g(t)$ and $f(t)$ do not overlap
- Area under their product is zero

Graphical Convolution Example

- Result of convolution (5 intervals of interest):

$$y(t) = f(t) * g(t) = \begin{cases} 0 & \text{for } t < -2 \\ -\frac{3}{2}t^2 + 6 & \text{for } -2 \leq t < 0 \\ 6 & \text{for } 0 \leq t < 2 \\ \frac{3}{2}t^2 - 12t + 24 & \text{for } 2 \leq t < 4 \\ 0 & \text{for } t \geq 4 \end{cases}$$

No Overlap
Partial Overlap
Complete Overlap
Partial Overlap
No Overlap



Convolution Demos

- **Johns Hopkins University Demonstrations**

<http://www.jhu.edu/~signals>

Convolution applet to animate convolution of simple signals and hand-sketched signals

Convolve two rectangular pulses of same width gives a triangle (see handout E)

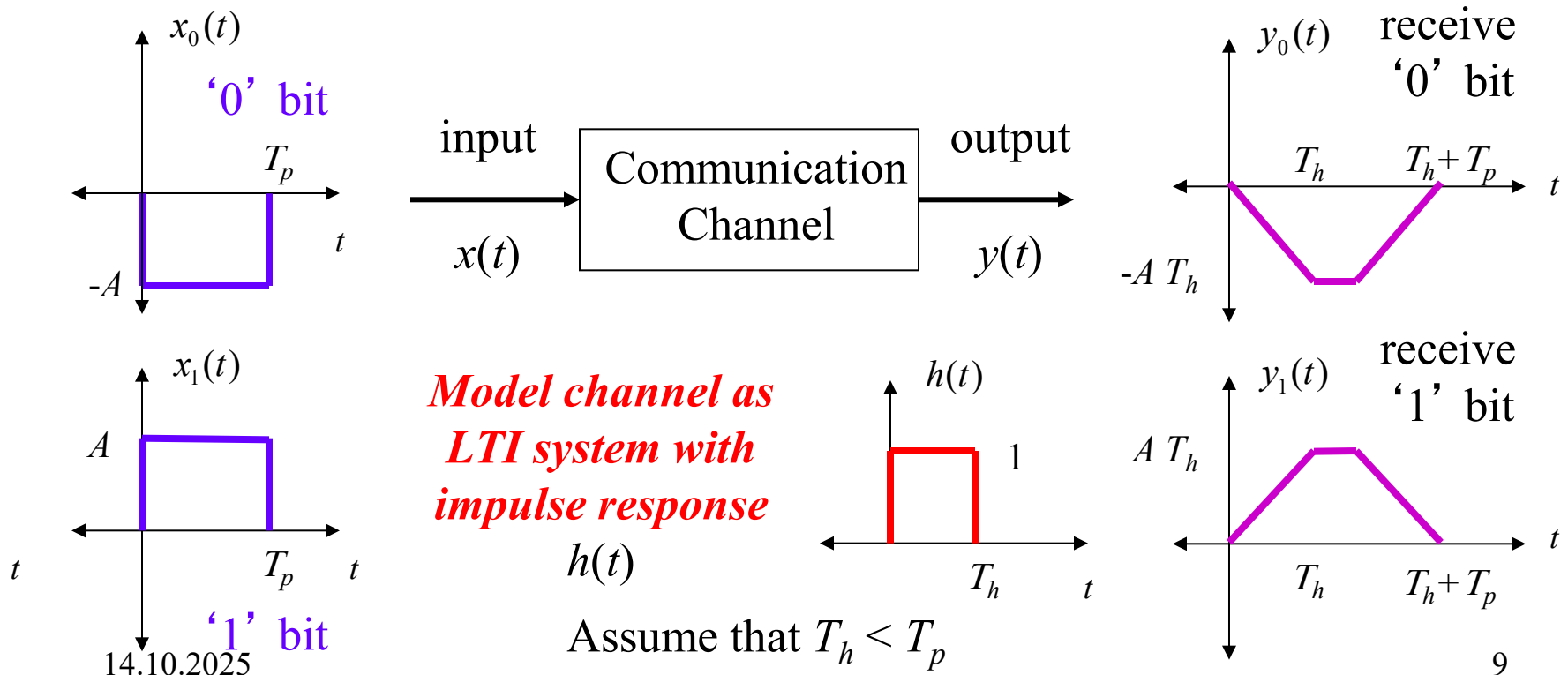
- **Some conclusions from the animations**

Convolution of two causal signals gives a causal result

Non-zero duration (called extent) of convolution is the sum of extents of the two signals being convolved

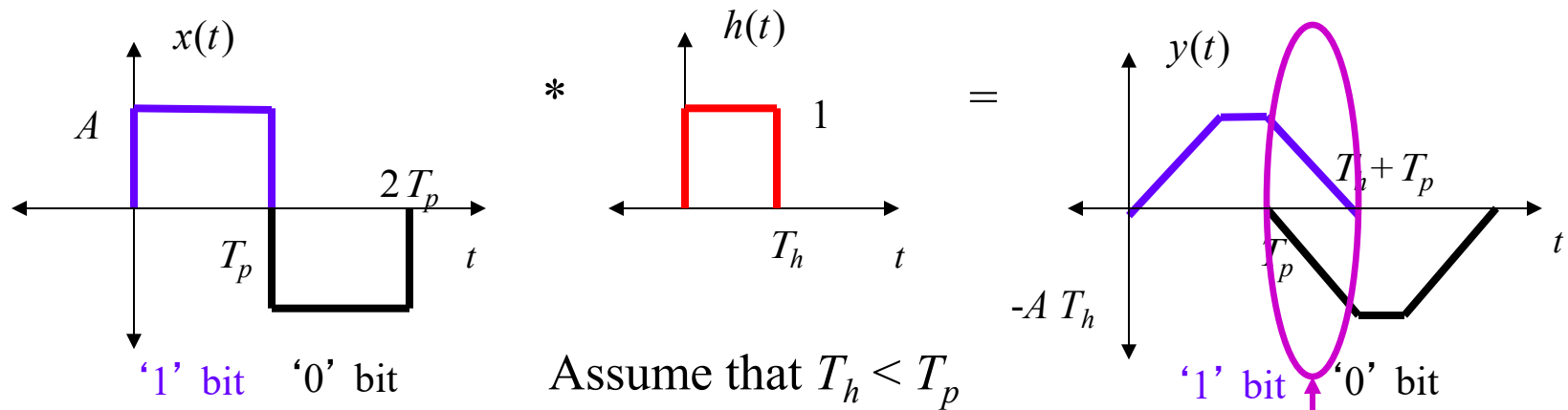
Example: Transmit One Bit

- Transmission over communication channel (e.g. telephone line) is analog



Example: Transmit Two Bits (Interference)

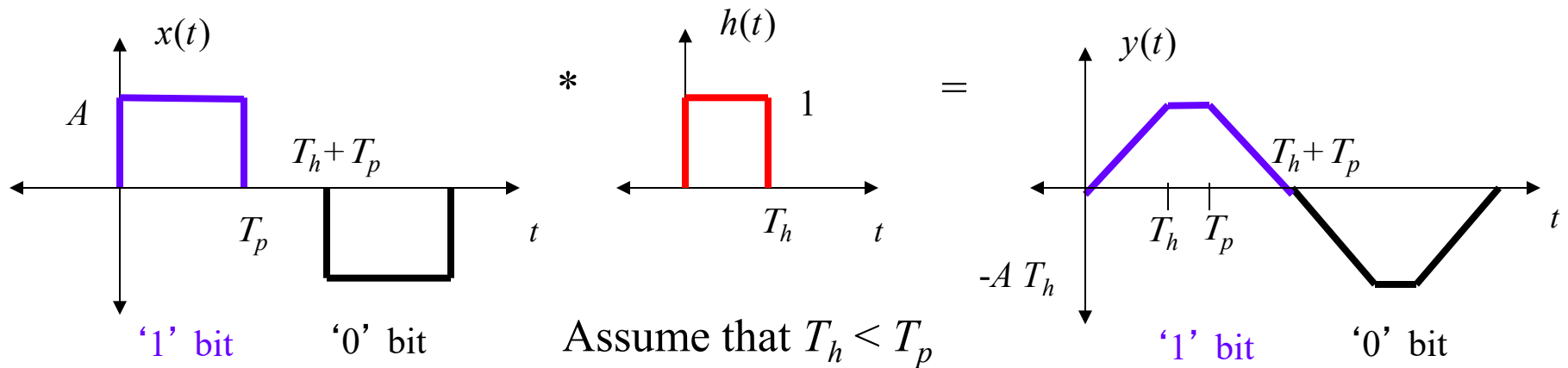
- Transmitting two bits (pulses) back-to-back will cause overlap (interference) at the receiver



- How do we prevent intersymbol interference at the receiver?

Example: Transmit Two Bits (No Interference)

- Prevent intersymbol interference by waiting T_h seconds between pulses (called a guard period)



- Disadvantages?